

Security risk assessment report

Part 1: Select up hardening tools and methods to implement

Eradicate Defaults: Immediate audit and change all default credentials across the infrastructure.

Enforce Strong Password Policies: Configure Network-level policies to require a minimum of 14 characters, complexity, and password history enforcement to prevent recycling.

Implement MFA: Mandate Multi-Factor Authentication for all users and employees within the organization.

Firewall Configuration: Configure firewall to filter inbound and outbound traffic

Setup IDPS: Implementation of an Intrusion detection and protection system as an extra layer of security and enabling real-time monitoring traffic and logs

Penetration testing: Initiate a comprehensive penetration to identify potential vulnerabilities in systems, networks, applications and operational processes.

Part 2: Explain your recommendations

To lock down the network, we should immediately eliminate all default credentials and enforce strict password rules, requiring 14 character complex passwords and blocking password recycling ensuring the organization stays safe from password guessing and brute force attacks. We should also mandate Multi-Factor Authentication (MFA) for every employee and gradually roll it out for users to ensure that a stolen password alone isn't enough to breach the system or compromise users' accounts. Additionally, we would need to reconfigure our firewalls to strictly filter both incoming and outgoing traffic, which blocks unauthorized access at the perimeter and prevents sensitive data from being exfiltrated if an account is compromised.

To provide an active layer of defense, I strongly recommend deploying an Intrusion Detection and Prevention System (IDPS) across our network perimeter. This system

will continuously monitor all network traffic in real-time to instantly identify and block known attack signatures or suspicious behavioral anomalies. By actively dropping malicious packets before they reach our internal systems, the IDPS acts as a critical safety net to stop future exploits in their tracks.

A comprehensive simulated penetration testing would help identify potential vulnerabilities in systems, networks, applications and operational processes to help the organization to better protect critical assets and customers personally identifiable information (PII).